

Warm-up

(1) Compute $\frac{d}{dx} \sin(x^2 + 1)$.

(2) Compute $\int \sin(x) + 5x^2 - 2 \, dx$.

Derivative rules

Compute the following derivatives:

$$\frac{d}{dx} (5x^7 - 3x) \left(x^{4/3} + \frac{1}{x} \right) =$$

$$\frac{d}{dx} \sec(\tan(\cos((x+1)^2))) =$$

$$\frac{d}{dx} \frac{\sin(\csc(x^2 + 1))}{x^4 + \cos(x)} =$$

Integration

Compute the following integrals:

$$\int \sin^4(t) \cos(t) \, dt =$$

$$\int \frac{x^2}{(x^3 - 3)^3} \, dx =$$

$$\int_0^2 \frac{x^2}{\sqrt{5x^3 + 9}} \, dx =$$